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MyDesign Portfolio

Element E Initial Template

STEM Principles and Practices to Design Requirements (E1)

The solution has to be able to grip tightly onto any drawstring without slipping or falling off

STEM Concept: Mathematics, Engineering

To archive this design requirement, we need to use math to be able to determine the circumference of the string and how big to make out clip as well. We also need to use engineering to build and design the prototype.

The solution has to be sturdy and not cause damage to anything it comes in contact with.

STEM Concept: Problem solving, testing

Because most the materials that we use today include plastic, metal, or glass, we need to think of other less common materials to use to make our project and design. We also need to go through trial and error to understand how each material reacts in the washing machines and find the material that works best.

Must make sure that the hoodie string does not get lost.

STEM Concept: Engineering, trial and error testing

To make sure that our prototype serves its purpose, we need to go through a lot of testing and reiterating our design to make sure that it is working.

Must look fashionable or cute to wear outside

STEM Concept: science, technology, engineering

We need to use technology to research and design the prototype and create different design for all people. We need to engineer and design pictures, characters, shapes, symbols to complement all fashion styles.

Must be something that can stay on effortlessly

STEM Concept: consideration of stakeholders and be convenient for the stakeholders

The entire purpose of designing something is to make something more convenient than it already is. Currently, the only solution to this problem is to tie the strings together. This is

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very convenient. Making something that stays on effortlessly without further thought is very important because it ensures the stakeholder is happy and will purchase the product.

Proposed Solution to STEM Principles and Practices (E2)

STEM Principle for Design Solution #1

STEM Concept: fulfill its purpose in helping people

our solution with the croc clip, will be able to hold the string in place as well as look fashionable. Because people are struggling with having the string not get lost or out of the hoodie or track pants, our solution connects to the principle of helping people.

STEM Principle for Design Solution #2

STEM Concept: sustainable and use of recycled materials

Using metal to bend into a clip or recycled plastic to 3d print a clip design is good for maintaining the environment and making good use of materials that others might throw out.

STEM Principle for Design Solution #3

STEM Concept: It is easy to use for everybody

Describe how STEM concept applies directly to the specifics of your solution: the velcro straps are adjustable and the velcro will always stick to one another, so it is perfect for different sized strings. That way, everyone can just purchase the same item and it will all work. The velcro design will be easy for people to understand how to work it and it will not require any advance education or readings to understand how to use our solution.

Evidence Design is Reviewed (E3)

Documentation #1: Mr. Sid

- Wondered why we choose this problem because he himself does not really have this problem
- Suggested we just tie a knot
- Mentioned how the dryer has high temperatures
- Mentioned how there might be a lot of sound if we use hard materials

Documentation #2: Mr. Steinhorn

- Agreed that using a crochet hook is annoying
- Suggested we make the hole containing the string smaller or bigger

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- Wondered how we would test, and suggested that we make sure the device does not get stuck or impact anything around it

Experts with Credentials (E4)*Expert #1:* Mr. Sid - IT person*Expert #2:* Mr. Steinhorn - electrical engineer**Verification or Corrective Response to Design Review (E5)**

Interview with Expert #1: Mr. Sid told us to remember that the dryer and washer machine has high temperatures, which informed our decision to reconsider using hot glue or a meltable plastic.

Interview with Expert #2: Mr. Steinhorn told us that the rotating of the machines might cause some noise which informed our decision to reconsider using a hard material that might scratch.